

Zhengkai Zhang

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EXPERIENCE

- **Meta Platforms, Inc.** Menlo Park, CA
Software Engineer (E6), Inference Optimization Feb. 2022 – Present

LLM Inference & Model Performance. Hands-on optimization of serving runtimes and low-precision model deployment.

- Reduced time per output token (TPOT) by 30–40% by enabling EAGLE3 speculative decoding with adaptive draft-length tuning in vLLM for Muse Spark; optimized serving on Blackwell and Hopper systems through prefill/decode disaggregation and prefix caching.
- Built a low-precision weight-quantization pipeline, reducing quantization time from 1 hour to 10 minutes for GLM-5.2-scale models through CPU parallelization; supported calibration-free quantization and selective higher-precision layers, with model quality validated on major benchmarks.

RecSys Inference Performance. Technical lead for a team optimizing PyTorch-based inference for advanced ads and feed ranking models.

- Owned the FX-based inference compiler and lowering stack; replaced graph-level model-partitioning rewrites with native PyTorch method definitions and extended FX tooling, cutting research-to-production time for new architectures from years to months and enabling coding agents to edit and validate partition definitions as Python code.
- Drove model–infrastructure co-design with modeling and kernel teams, including compiler and runtime integration of In-Kernel Broadcast Optimization (IKBO) to eliminate redundant user-embedding replication.
- Drove system-level inference optimizations, achieving a 10× feature-preprocessing speedup over the CPU baseline by offloading execution to GPU and fusing operators to reduce kernel-launch overhead.
- Scaled embedding serving beyond single-device memory limits and improved lookup throughput by sharding large embedding tables across multiple devices.
- Enabled model portability across NVIDIA GPUs, AMD GPUs, CPUs, and Meta’s MTIA accelerators, taking new inference backends from initial bring-up to production.
- Upstreamed inference optimizations to PyTorch, FBGEMM, TorchRec, and AITemplate.

- **Amazon** Seattle, WA
Software Development Engineer Jul. 2019 – Jan. 2022

- Led the AWS serverless redesign of the Warehouse Overcharge Protection System, expanding supported capacity from 400 to 30,000 warehouses per region (75×) while preserving backward compatibility; partnered with stakeholders and two engineers to launch worldwide in four months.
- Parallelized Warehouse Risk Reporter data processing and migrated its datastore from S3 to DynamoDB.
- Built the risk-data ingestion pipeline, filtering and consolidating inputs from APIs, S3, and DynamoDB; delivered self-service APIs for warehouse onboarding without developer intervention.
- Enabled continuous deployment with automated approval workflows, integration tests, and rollback alarms.

EDUCATION

- **M.S. in Electrical and Computer Engineering** Jan. 2018 – May 2019
Carnegie Mellon University Pittsburgh, PA
- **B.S. in Electrical Engineering** Aug. 2015 – Dec. 2017
Texas A&M University College Station, TX
- **B.S. in Electrical Engineering** Aug. 2013 – May 2015
University of Electronic Science and Technology of China Chengdu, China
Transferred to Texas A&M University after sophomore year.

SELECTED PUBLICATION AND TECHNICAL BLOG

- [1] Y. Chen, L. Luo, B. Zhang, J. Jiao, B. Li, H. Wang, T. Tang, A. Cai, Z. Shen, **Zhengkai Zhang**, et al., “ROCS: Request-Oriented Compute Sharing for Efficient Large-Scale Recommendation,” *arXiv:2607.27744*, Meta AI, 2026.
- [2] “SOLARIS: Speculative Offloading of Latent-bAsed Representation for Inference Scaling.” *SIGIR 2026, Industry Track*. Co-author: **Zhengkai Zhang**.
- [3] X. Chen, Y. Chen, Z. Chen, J. Herbach, J. Jiao, J. Leng, D. Li, G. Liu, L. Luo, GP Musumeci, M. Shao, E. Wen, R. Xiao, Y. Yetim, N. Yu, **Zhengkai Zhang**, “Meta Adaptive Ranking Model: Bending the Inference Scaling Curve to Serve LLM-Scale Models for Ads,” *Engineering at Meta*.
- [4] J. Jiao, B. Li, H. Yu, Y. Fang, **Zhengkai Zhang**, Z. Zhao, et al., “In-Kernel Broadcast Optimization: Co-Designing Kernels for RecSys Inference,” *PyTorch Blog*.
- [5] L. Fang, S. Deng, H. Jia, H. Li, I. Mitra, S. Qin, **Zhengkai Zhang**, Z. Zhao, Z. Zheng, “Building Highly Efficient Inference System for Recommenders Using PyTorch,” *PyTorch Blog*.

SKILLS

- **Programming languages:** Python, C/C++, CUDA, Java, Go, Bash
- **Tools and frameworks:** PyTorch, Triton, TorchRec, FBGEMM, AITemplate, vLLM, SGLang, Docker, AWS